Multispectral Remote Sensing Image Deblurring Using Auxiliary Band Gradient Information

Zhuangtianyu Liao, Wenyi Zhang, Qingwei Chu, Hao Ding, and Yuxin Hu

***Abstract*— Multispectral remote sensing images (RSI), including hyperspectral and multispectral images, contain adequate information of ground objects and areas and play important roles in environmental monitoring, weather forecasting, urban planning, and so on. However, due to many inevitable external effects on the remote sensing pathway, RSIs are often degraded by blur. The fields of multispectral RSI deblurring have witnessed great improvements in recent years, including both optimization-based and deep-learning-based methods. However, issues are to be addressed in the RSI deblurring field, such as the incompatibility of general regularizations, lack of spectral correlations for multispectral RSIs, demands of blind deblurring for real-world applications, and high costs of computation. To address these problems, we incorporate a novel prior exploiting gradient information similarity between different spectral bands, and we name it auxiliary band gradient information (ABGI) prior. We show that the ABGI prior is applicable to all gradient sparsity regularizations by a simple subtract-then-add step. Specifically, we apply ABGI prior to the patchwise minimal pixel (PMP) prior-based deblurring method, and we also prove that the PMP prior exhibits sparsity for clear natural RSIs. We estimate our method on RSI datasets of different spectral types and geographic resolutions. Compared with other state-of-the-art deblurring methods, our method shows superior performance on both simulated and real-world blur.**

***Index Terms*— Auxiliary band gradient information (ABGI), blind deblurring, remote sensing image (RSI).**

I. Introduction

REMOTE sensing images (RSI) are mostly acquired from the optical camera or the synthetic aperture radar (SAR) carried by airplanes or satellites. The acquisition of RSIs often suffers various degradation, such as noises (including Gaussian random noise, impulse noise, and so on), striping, blurring, or their complex combination. RSI restoration aims at removing degradations from RSIs and recovering naturalness and clearness. RSI deblurring focuses on recovering ground objects from various blur effects. Many inevitable factors during the acquisition of RSIs, e.g., atmospheric disturbance [1], camera defocus [2], platform high-frequency jitters [3], and sensor instability [4], are some possible causes of blur [5]. Unlike sparse noises, blur effects are usually equivalent to low-pass filters, remaining low-frequency components, such as rough contours of objects, and filtering high-frequency components, such as edges and textures. Blurred RSIs lose not only visual clearness for human eye observation but also crucial information of ground objects, causing inferior performance of subsequent applications, such as object detection [6], object recognition [7], object tracking [8], semantic segmentation [9], change detection [10], [11], [12], [13], and so on. Therefore, research on RSI restoration is significant.

The development of image deblurring methods boosts their application in the fields of remote sensing. To date, RSI deblurring methods have made significant improvements in recovering quality, including both blind [14], [15] (i.e., blur kernel is not known in advance) and nonblind methods [16], [17], [18], [19], [20], [21]. However, there are still many issues to be addressed.

1) General regularizations are not completely compatible with multispectral or hyperspectral RSIs. Current deblurring methods mostly use global regularizations, which reflect global yet neglect local constraints. For example, intensity gradient sparsity prior assumes that the intensity gradient of an image is sparse in total. However, the assumption is not always satisfied when the clear ground truth includes adequate details, and it is ignorant of the gradient spatial distribution.
2) Hyperspectral images are more often discussed for image restoration tasks, since it has adequate spectral information, while multispectral images lack discussions. Hyperspectral images have higher spectral resolution and more bands; thus, spectral correlations (e.g., low-rank property [20], [21]) can be exploited as effective priors. It is harder to exploit spectral information from multispectral images, and the hyperspectral methods are not fully compatible with multispectral images.
3) Nonblind RSI deblurring methods are discussed much more than blind ones currently, which is not consistent with real-world application requirements. Blur effects in the real world are random and unpredictable; thus, knowing the blur kernel in advance is almost impossible. Blind deblurring is a must for real-world applications

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despite its high difficulty and few discussions in the remote sensing fields.

4) Current image deblurring methods are mostly computationally demanding. Sophisticatedly designed regularizations usually require tremendous computational cost to recover a small-size hand-taken image. However, the scale of RSIs is far beyond that of hand-taken images, and the computational cost will be unacceptable. If new regularizations are to be introduced, they should be as computationally friendly as possible to cope with large-scale RSIs.

To alleviate the abovementioned problems, we propose a novel prior named auxiliary band gradient information (ABGI) for multispectral RSI deblurring. The ABGI prior assumes the clear latent of blurred band has high gradient similarity with clear bands. We apply it on an improved method based on the blind image deblurring algorithm using patchwise minimal pixels (PMPs) [22]. The main contributions of the proposed method are as follows.

1) We observe that different bands from the same image show high gradient similarity, and use the clear band aligned with the blurred band as an extra gradient position prior, calling it ABGI prior.
2) We propose a blind multispectral RSIs deblurring method that incorporates ABGI prior. We validate the effectiveness of the ABGI prior on high-resolution RSIs. We also show that the ABGI prior does not depend on specific regularization form, is applicable to all gradient sparsity regularizations by a simple subtract-then-add step, and is, therefore, computationally friendly.
3) We evaluate our method on both open RSI datasets and the latest GaoFenDuoMo (GFDM; meaning high-resolution multimodal) satellite RSIs and compare it with state-of-the-art blind deblurring methods. Our quantitative and qualitative results show superior deblurring performance on both simulated and real-world scenes. Moreover, we show by the experimental results that our method fits different spectral types (hyperspectral and multispectral) and geographic resolutions (both high and medium).

The remainder of this article is organized as follows. Section II introduces related works on the fields of image deblurring and its application on remote sensing. Section III introduces PMP and ABGI priors and proves the sparsity of PMP and the correctness of ABGI prior utilization by a simple subtract-then-add step. Section IV shows the overall procedure and the detailed algorithm. Section V evaluates the quantitative and qualitative performance of the proposed method on simulated and real blur and compares it with some state-of-the-art blind image deblurring methods. Section VI is the analysis part of our proposed method. Section VII gives the conclusion of this article.

II. Related Work

A. Image Deblurring

The image deblurring frameworks are classified into optimization-based and deep-learning-based frameworks. Optimization-based frameworks solve the deblurring problem by modeling a minimization problem. Deep-learning-based frameworks introduce high-level feature extractors or classifiers to assist deblurring.

1) Optimization-Based Methods: The maximum-a posteriori (MAP)-based deblurring methods are most widely used due to their effectiveness and stability, and many previous works have proved this. With proper regularizations, MAP-based methods are capable to avoid being ill-posed and recover nontrivial blur kernel [23], and the kernel estimation procedure is proved to be convergent [24]. It is proved that if the regularizations are nonconvex or nonsmooth, the problem can still be efficiently solved under sufficient conditions or smoothing strategies [25].

Besides theoretical analyses of MAP-based methods, lots of previous works have exploited variant image priors for deblurring. Dark channel prior (DCP) is proposed by He et al. [26] and applied on image deblurring by Pan et al. [27], which exploits the sparsity of dark pixels on clear images. Extreme channel prior (ECP) [28] is a similar approach but it exploits the sparsity of both bright and dark pixels. Other works include normalized prior [29], local maximum gradient prior [30], latent structure prior [31], and so on.

2) Deep-Learning-Based Methods: Deep-learning-based frameworks are gathering increasing attention in recent image deblurring works. Xu et al. [32] first use the deep neural network to capture the characteristics of image degradation and alleviate artifacts. Xu et al. [33] propose a novel deep convolutional neural network (CNN) structure to extract edges from blurred images and dramatically simplify kernel estimation. Gong et al. [34] incorporate CNN to estimate the motion flow from motion-blurred images and recover them. Nah et al. [35] make an end-to-end approach to recover clear images from various blur effects via a multiscale CNN. Generative adversarial network (GAN) approaches [36], [37] also boost the development of deep-learning-based deblurring. Kupyn et al. [36] present an end-to-end GAN to remove motion blur from images. Its later version [37] incorporates various backbones to fit different performance or quality requirements. Besides the direct use of neural networks, deep-learning-based image priors [38], [39], [40] also show excellent performance.

B. RSI Deblurring

Image deblurring methods have been applied in many interdisciplinary fields, including remote sensing [3], [5], [14], [15], [41], [42], [43], [44]. The earliest work may trace back to [45], which recovers blurred RSIs from highly incomplete measurements. Both MAP-based and deep-learning-based approaches are continuously proposed. Many methods incorporate general regularizations to deblur RSIs. Manwei et al. [41] propose a method combining DCP and joint bilateral filtering to enhance the clearness of edges and textures. Lim et al. [43] make a fusion of L_0-norm and L_2-norm regularization to address the problem of texture loss. Zhang et al. [44] combine deblurring and destriping of on-orbit RSIs using gradient sparsity property. Han et al. [5] combine extreme value sparsity priors as regularizations.

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Specific regularizations for RSIs are also frequently discussed. Hyperspectral images contain adequate spectral and spatial information and are, therefore, under frequent discussion [45]. Henrot et al. [17] perform nonblind deconvolution on hyperspectral images using a positive spatial and spectral smoothness regularization. Fang et al. [19] introduce a variation of total variation regularization to exploit both spatial and spectral correlations. Chang et al. [20] propose a unified low-rank tensor recovery model to exploit spectral and spatial similarities. Based on this model, [21] exploits the low-rank property in both images and sparse error components and becomes robust to complicated degradations. Li et al. [46] use the spectral channel correlations to derive a reference image to help restore defocused hyperspectral images.

Deep-learning-based approaches in recent years seem to prefer GAN more than other deep models. Wang et al. [3] use a discriminative network-based classifier of clearness as a regularization of latent estimation. Xia and Liu [42] propose a Wasserstein generative adversarial network (WGAN)-based end-to-end model to recover RSIs from blur. Fang et al. [47] consider deblurring as a domain transformation problem and combine image restoration and image texture complexity recognition subtasks to implement RSI motion deblurring. Li et al. [48] propose an approach, which registers both the clear reference image and the blurred image into multilevel feature space, and use clear texture features to assist the deblurring.

III. PMPs and ABGI

A. Notations and Preliminaries

Assuming that the blurring process of a single image is uniform and spatially invariant, the image deblurring problem is modeled as follows:

$$\mathbf{Y} = \mathbf{X} \otimes \mathbf{K} + \mathbf{N} \tag{1}$$

where $\mathbf{Y}$ denotes the blurred image, $\mathbf{X}$ denotes the clear latent image, $\otimes$ represents convolution operation, $\mathbf{K}$ denotes the blur kernel, and $\mathbf{N}$ denotes additive noise.

The optimization-based blind image deblurring method estimates the blur kernel and clear latent image simultaneously by optimizing the following problem:

$$\min_{\mathbf{X},\mathbf{K}} (\mathcal{L}(\mathbf{X} \otimes \mathbf{K}, \mathbf{Y}) + \lambda \mathrm{T}(\mathbf{K}) + \mu R(\mathbf{X})) \tag{2}$$

where $\mathcal{L}(\cdot,\cdot)$ represents data fidelity term, which measures a distance between $\mathbf{Y}$ and $\mathbf{X}\otimes\mathbf{K}$ and enforces them to be consistent. $\mathrm{T}(\mathbf{K})$ denotes the regularization term corresponding to the blur kernel $\mathbf{K}$, $R(\mathbf{X})$ denotes regularization term corresponding to $\mathbf{X}$, and μ and λ are nonnegative parameters, balancing the data fidelity term and each regularization terms in the optimization process. Practically, the Frobenius norm is chosen as the data fidelity term if the additive noise term is assumed to be Gaussian; therefore, the key to recover high-quality latent images is to choose appropriate latent regularization terms.

A classical image deblurring method involves choosing appropriate image priors and constraints as regularization terms and optimizing the problem under the MAP framework.

To minimize (2), an iterative method optimizing $\mathbf{X}$ and $\mathbf{K}$ alternatively can be applied to obtain the solution. The optimizing problem can be divided into two subproblems as the following $\mathbf{X}$ problem and $\mathbf{K}$ problem:

$$\min_{\mathbf{X}} \left(\|\mathbf{Y} - \mathbf{X} \otimes \mathbf{K}\|_F^2 + \mu R(\mathbf{X}) \right) \tag{3}$$

$$\min_{\mathbf{K}} \left(\|\nabla\mathbf{Y} - \nabla\mathbf{X} \otimes \mathbf{K}\|_F^2 + \lambda \|\mathbf{K}\|_F^2 \right) \tag{4}$$

where $\|\cdot\|_F$ denotes matrix Frobenius norm and ∇ denotes gradient operator. Note that obtaining optimal kernel from loss of gradient yields more accurate result than from loss of intensity, so (4) substitutes intensity terms with gradient terms in its data fidelity term.

After the two-step optimization, both the blur kernel and intermediate latent image are obtained. However, the latent image obtained at this stage usually contains only simple textures, and it performs inferior when the images, particularly RSIs, record complicated backgrounds and textures [49]. The nonblind deconvolution step using the estimated blur kernel is consequently introduced to obtain the final deblurring result, which is usually modeled as the following optimization problem:

$$\min_{\mathbf{X}} \left(\left\|\mathbf{Y} - \mathbf{X} \otimes \hat{\mathbf{K}}\right\|_F^2 + \mu_R \|\nabla\mathbf{X}\|_R \right) \tag{5}$$

where $\hat{\mathbf{K}}$ denotes the estimated kernel, μ_R is a positive parameter, $\|\cdot\|_R$ denotes a norm imposing sparsity constraint on the gradient, e.g., L_0-norm, the number of nonzero elements of a vector or matrix, or L_1-norm or their combination is applied here. Imposing such a penalty on gradient ensures sparsity of image gradient, which is often the case for most of the natural clear images.

B. Patchwise Minimal Pixels

PMP is a simple and effective local minimal intensity prior proposed by Wen et al. [22]. Let $\mathbf{X} \in [0,1]^{m\times n\times c}$ denote an image with size of $m \times n$, c channels, and normalized pixel intensity. The image x can be divided into small patches with size s_p, each pair of which has no overlapping, so in total the number of patches is $P = \lceil \frac{m}{s_\mathrm{p}} \rceil \times \lceil \frac{n}{s_\mathrm{p}} \rceil$. Let $\mathcal{P}_i$ denote the index set of pixels in the ith patch, where $i \in \{1, 2, \ldots, P\}$. The PMP of $\mathcal{P}_i$ is defined as the pixel, which has minimal intensity among all pixels in the ith patch, i.e.,

$$\mathrm{PMP}(\mathbf{X}, \mathcal{P}_i) = \min_{(x,y)\in\mathcal{P}_i} \min_{c\in\mathcal{C}} \mathbf{X}(x, y, c) \tag{6}$$

where (x, y) is the pixel coordinate, $\mathcal{C}$ denotes all channels of the image, and $\mathbf{X}(x, y, c)$ denotes the pixel intensity at position (x, y) and channel c. Moreover, we define the PMP of an image $\mathbf{X}$ a function mapping $[0, 1]^{m\times n\times c}$ to $[0, 1]^P$

$$\mathrm{PMP}(\mathbf{X}) = [\mathrm{PMP}(\mathbf{X}, \mathcal{P}_1), \ldots, \mathrm{PMP}(\mathbf{X}, \mathcal{P}_P)]^\mathrm{T}. \tag{7}$$

It is developed from DCP proposed by He et al. [26] and applied as an image deblurring prior by Pan et al. [27]. Similarly, the DCP of a pixel and the DCP of an image $\mathbf{X}$ are defined as follows:

$$\mathrm{DCP}(\mathbf{X}, x, y) = \min_{(x',y')\in\mathcal{P}(x,y)} \min_{c\in\mathcal{C}} \mathbf{X}(x', y', c) \tag{8}$$

$$\mathrm{DCP}(\mathbf{X}) = \left[\mathrm{DCP}(\mathbf{X}, x_1, y_1), \ldots, \mathrm{DCP}(\mathbf{X}, x_{m\times n}, y_{m\times n})\right]^\mathrm{T}. \tag{9}$$

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The definition of DCP is almost the same as PMP, and PMP is actually a subset of DCP. One important difference between DCP and PMP is the selection of patches. Patches for DCP are selected from every pixel, resulting in totally $m \times n$ overlapping patches in DCP representation and computation, while PMP divides the image into size-fixed and nonoverlapping patches. With an appropriate value of patch size, PMP requires much less computational cost than DCP without loss of sparsity as an image regularization.

Similar to DCP, the motivation of PMP is consequently the sparsity of image local minima. Intuitively, the PMP values of clear natural images are usually less than that of blurred images, because blurring is equivalent to low-pass filtering and has the effect of averaging. When a clear image becomes blurred, its minimal pixels are nondecreasing and likely to obtain greater values. PMP shows sparsity especially when the minimal pixels in clear images take zero.

Although the sparsity of DCP is proved by Pan et al. [27] and PMP is a subset of DCP, proof of DCP is not applicable for PMP. DCP is a pixel-based selection, and the sparsity relationship can be reduced by correspondence from pixels to patches naturally. Nevertheless, PMP is a patch-based selection, and we must construct the correspondence from patches to some representative pixels to reduce the sparsity relationship, which is nontrivial. To the best of our knowledge, most of the methods incorporating PMP verify PMP sparsity statistically, but few of them have theoretical discussions. In our paper, we prove the fact that the PMP of blurred images is less sparse than that of clear images under the following assumption.

Assumption 1: The valid size of the blur kernel is no greater than that of PMP patches.

The valid size here refers to the possible minimal size of a shrank blur kernel, such that no nonzero coefficients are lost. Assumption 1 is based on the observations that natural blur kernels are usually sparse and concentrate nonzero coefficients at or around the kernel center, so the valid size of a blur kernel is usually small enough despite the size of the whole kernel. In our experiments, the patch size is computed based on the height and width of the image and is usually enough to satisfy Assumption 1.

Channels are mutually independent during the convolution process, so we consider the single-channel case without loss of generality in the following discussion. Let $\mathbf{Y}$ denote the image blurred by convoluting a clear image $\mathbf{X}$ with a normalized nonnegative kernel $\mathbf{K}$, i.e., $\sum_{i,j} \mathbf{K}(i, j) = 1$ and $\mathbf{K}(i, j) \geq 0$. Based on Assumption 1, we introduce the following lemma first.

Lemma 1: Let (x, y) denote a pixel index of $\mathbf{X}$ or $\mathbf{Y}$. If Assumption 1 is satisfied, then the following inequality holds:

$$\mathrm{DCP}(\mathbf{Y}, x, y) \geq \mathrm{DCP}(\mathbf{X}, x, y). \tag{10}$$

The proof of Lemma 1 is provided in [27, Supplementary Materials]. Based on Lemma 1, we have the following property.

Property 1: If Assumption 1 is satisfied, then the following inequality holds:

$$\mathrm{PMP}(\mathbf{Y}, \mathcal{P}_i) \geq \mathrm{PMP}(\mathbf{X}, \mathcal{P}_i). \tag{11}$$

Proof: [Proof of Property 1] The key to prove Property 1 is that every patch has a pixel $(x_i, y_i) \in \mathcal{P}_i$, such that $\mathrm{DCP}(\mathbf{Y}, x_i, y_i) = \mathrm{PMP}(\mathbf{Y}, \mathcal{P}_i)$. This fact holds because of Assumption 1, and it allows us to find the satisfying pixel. Specifically, we choose the center pixel (x_i, y_i) of both patch $\mathcal{P}_i$ and the kernel. In the case where the kernel and the patch have the same size, it is apparent that both $\mathrm{DCP}(\mathbf{Y}, x_i, y_i) = \mathrm{PMP}(\mathbf{Y}, \mathcal{P}_i)$ and $\mathrm{DCP}(\mathbf{X}, x_i, y_i) = \mathrm{PMP}(\mathbf{X}, \mathcal{P}_i)$ hold from the definition; in the case which the patch has rigorously greater size, we pad the kernel with zeros and extend it to the same size as the kernel; then, it is reduced to the first case. Therefore, we obtain $\mathrm{DCP}(\mathbf{Y}, x_i, y_i) = \mathrm{PMP}(\mathbf{Y}, \mathcal{P}_i)$ and $\mathrm{DCP}(\mathbf{X}, x_i, y_i) = \mathrm{PMP}(\mathbf{X}, \mathcal{P}_i)$. By these equalities, we directly obtain $\mathrm{PMP}(\mathbf{Y}, \mathcal{P}_i) \geq \mathrm{PMP}(\mathbf{X}, \mathcal{P}_i)$ from Lemma 1. ■

Since image intensity is nonnegative, PMP values are all nonnegative, so we can directly obtain the following corollary.

Corollary 1: The following inequality holds:

$$\|\mathrm{PMP}(\mathbf{Y})\|_R \geq \|\mathrm{PMP}(\mathbf{X})\|_R. \tag{12}$$

Thus, it is shown that clear images have sparser PMP than blurred images, and PMP is an efficient image prior for deblurring.

C. Auxiliary Band Gradient Information

Most image deblurring scenes are completely blind, and we need to solve the ill-posed deblurring problem to estimate both latent image and kernel using various priors. For gradient sparsity prior, the most frequently used prior for image deblurring, it exploits gradient sparsity from clear natural images and imposes such a property on the deblurring process. These priors are mostly assumed based on all nature images and, thus, show generality. However, deblurring special images need not only general assumptions but also special constraints as priors.

An RSI usually records the intensity of electromagnetic waves with different wavelength ranges within a fixed region on earth. These wavelength ranges are also known as bands in remote sensing fields. Different spectral bands of the same RSI are intrinsically aligned, since they are collected simultaneously by cameras or sensors, and a single image pixel contains multispectral intensity information of its real position. The reflection of different substances on earth is also different, so the intensity of different bands usually differs. However, gradients of different clear bands from the same image reflect the same textures or edges of ground objects and show spatial distribution similarity, so we assume that the latent image to recover is also similar to known bands in the aspect of gradient spatial distribution. Fig. 1 shows that though the spectral bands differ, their gradient maps are similar. We call these known bands "auxiliary bands," expecting these bands to be good references of gradient spatial distribution, and they are introduced in our proposed deblurring framework as

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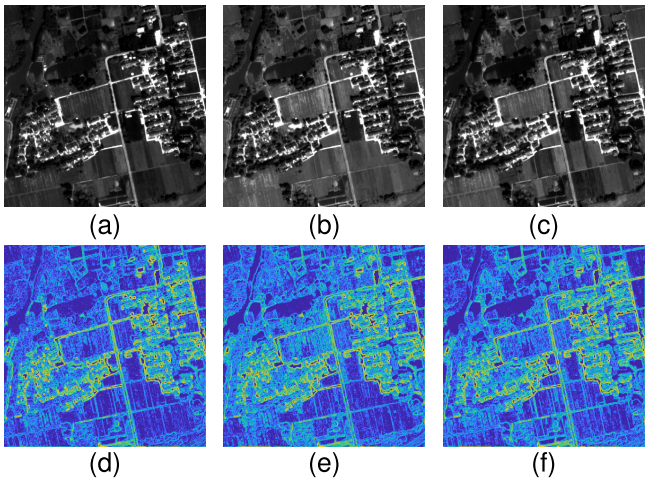


Fig. 1. Example of spectral band similarity from GFDM images. (a) Band 1. (b) Band 2. (c) Band 3. (d) Intensity gradient map of band 1. (e) Intensity gradient map of band 2. (f) Intensity gradient map of band 3.

ABGI prior. The deconvolution objective using ABGI prior regularization term is shown as (13), substituting gradient term with gradient difference. $\hat{\mathbf{K}}$ here denotes the estimated kernel. Note that if all pixels of the auxiliary band take the same values, the auxiliary gradient becomes zeros. In this case, ABGI regularization is degraded to intensity gradient sparsity prior, and the objective is degraded to that of the PMP-based method. We observe that the gradient difference of two clear bands from the same image also shows sparsity, and experimental details will be discussed in Section VI

$$\min_{\mathbf{X}}\left(\left\|\mathbf{Y}-\mathbf{X}\otimes\hat{\mathbf{K}}\right\|_F^2+\mu_R\|\nabla\mathbf{X}-\nabla\mathbf{X}_A\|_R\right). \tag{13}$$

Since different bands of the same RSI are spatially aligned, it is rather easy to utilize ABGI prior by a subtract-then-add approach during deconvolution. Before deconvolution, we subtract the blurred auxiliary band, and after the deconvolution, we add the clear auxiliary band back. Specifically, we blur the auxiliary band $\mathbf{X}_A$ with the estimated kernel $\hat{\mathbf{K}}$ and obtain blurred auxiliary band $\mathbf{Y}_A=\mathbf{X}_A\otimes\hat{\mathbf{K}}$, and then make difference $\mathbf{Y}_D$ with the blurred image $\mathbf{Y}$, i.e., $\mathbf{Y}_D=\mathbf{Y}-\mathbf{Y}_A\otimes\hat{\mathbf{K}}$. Since the convolution operator is linear, we assume that this equality also holds for the latent image, i.e., $\mathbf{X}_D=\mathbf{X}-\mathbf{X}_A$. Then, the deconvolution is performed on $\mathbf{Y}_D$, yielding $\hat{\mathbf{X}}_D$. The final estimation of latent image $\hat{\mathbf{X}}$ is then the sum of $\hat{\mathbf{X}}_D$ and $\mathbf{X}_A$. This property can be formulated as follows.

Property 2: Let $\hat{\mathbf{X}}$ and $\hat{\mathbf{X}}_D$ denote the optimal solution to (5) with respect to $\mathbf{X}$ and $\mathbf{X}_D$

$$\hat{\mathbf{X}}=\arg\min_{\mathbf{X}}\left(\left\|\mathbf{Y}-\mathbf{X}\otimes\hat{\mathbf{K}}\right\|_F^2+\mu_R\|\nabla\mathbf{X}-\nabla\mathbf{X}_A\|_R\right) \tag{14}$$

$$\hat{\mathbf{X}}_D=\arg\min_{\mathbf{X}_D}\left(\left\|\mathbf{Y}_D-\mathbf{X}_D\otimes\hat{\mathbf{K}}\right\|_F^2+\mu_R\|\nabla\mathbf{X}_D\|_R\right) \tag{15}$$

then, the following equality holds:

$$\hat{\mathbf{X}}=\hat{\mathbf{X}}_D+\mathbf{X}_A. \tag{16}$$

Proof: [Proof of Property 2] Utilizing $\mathbf{Y}_D=\mathbf{Y}-\mathbf{Y}_A$, $\mathbf{X}_D=\mathbf{X}-\mathbf{X}_A$, and $\mathbf{Y}_A=\mathbf{X}_A\otimes\hat{\mathbf{K}}$, we obtain

$$\begin{aligned}\hat{\mathbf{X}}_D&=\arg\min_{\mathbf{X}_D}\left(\left\|\mathbf{Y}_D-\mathbf{X}_D\otimes\hat{\mathbf{K}}\right\|_F^2+\mu_R\|\nabla\mathbf{X}_D\|_R\right)\\&=\arg\min_{\mathbf{X}}(\|\mathbf{Y}-\mathbf{Y}_A-(\mathbf{X}-\mathbf{X}_A)\\&\qquad\otimes\hat{\mathbf{K}}\|_F^2+\mu_R\|\nabla\mathbf{X}-\nabla\mathbf{X}_A\|_R)-\mathbf{X}_A\\&=\arg\min_{\mathbf{X}}\left(\left\|\mathbf{Y}-\mathbf{X}\otimes\hat{\mathbf{K}}\right\|_F^2+\mu_R\|\nabla\mathbf{X}-\nabla\mathbf{X}_A\|_R\right)-\mathbf{X}_A\\&=\hat{\mathbf{X}}-\mathbf{X}_A.\end{aligned} \tag{17}$$

■

This property shows that solving (14) is equivalent to solving the proximal objective (15). With the conclusion of Property 2, our subtract-then-add approach is proved to impose ABGI during deconvolution. This property is not only computationally convenient but also independent of choices of regularization norms $\|\cdot\|_R$, so we can easily apply ABGI on many current regularization terms.

IV. Image Deblurring Algorithm With PMP and ABGI

With PMP and ABGI priors, we propose the following deblurring procedure mainly based on the improved deblurring algorithm by Wen et al. [22]. Two main steps are involved in this procedure: kernel estimation and nonblind deconvolution.

During the kernel estimation step, gradient sparsity and PMP prior are imposed, and the objective (2) is rewritten as follows:

$$\begin{aligned}\min_{\mathbf{X},\mathbf{K}}\big(&\|\mathbf{Y}-\mathbf{X}\otimes\mathbf{K}\|_F^2+\lambda\|\mathbf{K}\|_F^2\\&+\mu\|\nabla\mathbf{X}-\nabla\mathbf{X}_A\|_0+\gamma\|\mathrm{PMP}(\mathbf{X})\|_0\big)\end{aligned} \tag{18}$$

which is divided into an alternative optimization with respect to intermediate latent image $\mathbf{X}$ and blur kernel $\mathbf{K}$

$$\min_{\mathbf{X}}\big(\|\mathbf{Y}-\mathbf{X}\otimes\mathbf{K}\|_F^2+\mu\|\nabla\mathbf{X}-\nabla\mathbf{X}_A\|_0+\gamma\|\mathrm{PMP}(\mathbf{X})\|_0\big) \tag{19}$$

$$\min_{\mathbf{K}}\big(\|\nabla\mathbf{Y}-\nabla\mathbf{X}\otimes\mathbf{K}\|_F^2+\lambda\|\mathbf{K}\|_F^2\big). \tag{20}$$

A. Initialization

The two-stage optimization starts with appropriate initial value $\mathbf{X}_0$ and $\mathbf{K}_0$, and the solution in the ith iteration $\mathbf{X}_i$ and $\mathbf{K}_i$ is solved using $\mathbf{X}_i$ and $\mathbf{K}_i$. Initialization of blur kernel adopts a coarse-to-fine search [50]. Each iteration starts with its last result, preventing it from obtaining a trivial delta solution. The original blurred image is used as initialization of latent image, i.e., $\mathbf{X}_0=\mathbf{Y}$.

B. Latent Image Estimation

The objective (19) containing L_0-norm of ABGI and PMP term is nonconvex and difficult to optimize directly. For the PMP term, we adopt a thresholding approach at the beginning of each iteration to ensure the PMP value is nonincreasing. This approximate approach avoids involving the PMP term explicitly in the objective. For the ABGI term, the introduction of auxiliary variables is frequently used for obtaining

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an approximate solution. Specifically, $\mathbf{Z}$ is introduced as an auxiliary variable, and the problem is approximated as follows, using large parameter ν to enforce $\|\nabla\mathbf{X} - \nabla\mathbf{X}_A - \mathbf{Z}\|_F \approx 0$. Thus, $\nabla\mathbf{X} - \nabla\mathbf{X}_A \approx \mathbf{Z}$

$$\min_{\mathbf{X}} \left(\|\mathbf{Y} - \mathbf{X} \otimes \mathbf{K}\|_F^2 + \nu \|\nabla\mathbf{X} - \nabla\mathbf{X}_A - \mathbf{Z}\|_F^2 + \mu \|\mathbf{Z}\|_0 \right). \tag{21}$$

For each outer iteration, we repeat utilizing PMP prior, estimating the auxiliary variable and updating latent image estimation for T times in the inner loop to reach convergence. Let $\mathbf{X}_i^t$ denote the latent image estimation at the ith outer iteration and the tth inner iteration, and $\mathbf{M}_i^t$ denote the PMP binary mask of $\mathbf{X}_i^t$, in which the element at pixel (x, y) takes 1 if the pixel is a PMP and otherwise 0. Specifically, let $\mathcal{S}(\mathbf{X}_i^t)$ denote the index set of PMPs of $\mathbf{X}_i^t$; then, the PMP mask is calculated as follows:

$$\mathbf{M}_i^t(x, y) = \begin{cases} 1, & (x, y) \in \mathcal{S}\left(\mathbf{X}_i^t\right) \\ 0, & \text{otherwise.} \end{cases} \tag{22}$$

For each repeat, we first update the latent image by the following equation:

$$\tilde{\mathbf{X}}_i^t = \mathbf{X}_i^t \odot \left(\mathbf{1} - \mathbf{M}_i^t\right) + \text{Thrshd}(\mathbf{X}_i^t \odot \mathbf{M}_i^t, \kappa) \tag{23}$$

where $\odot$ denotes elementwise multiplication of two same-size matrices, and the threshold function $\text{Thrshd}(\mathbf{X}, \kappa)$ keeps matrix elements whose absolute values are no less than κ and, otherwise, converts them to zero

$$\text{Thrshd}(\mathbf{X}, \kappa)(x, y) = \begin{cases} 0, & |\mathbf{X}(x, y)| < \kappa \\ \mathbf{X}(x, y), & \text{otherwise.} \end{cases} \tag{24}$$

Then, the estimation of auxiliary variable $\mathbf{Z}_i^t$ can be obtained by optimizing the following objective:

$$\min_{\mathbf{Z}_i^t} \left(\nu \left\| \nabla\tilde{\mathbf{X}}_i^t - \nabla\mathbf{X}_A - \mathbf{Z}_i^t \right\|_F^2 + \mu \left\| \mathbf{Z}_i^t \right\|_0 \right). \tag{25}$$

Note that the objective can be optimized elementwise, and its solution is given by [25]

$$\mathbf{Z}_i^t = \text{Thrshd}(\nabla\tilde{\mathbf{X}}_i^t - \nabla\mathbf{X}_A, \sqrt{\mu/\nu}). \tag{26}$$

Eventually, we update the latent image by optimizing the following objective:

$$\min_{\mathbf{X}_i^{t+1}} \left(\left\| \mathbf{Y} - \mathbf{K}_i \otimes \mathbf{X}_i^{t+1} \right\|_F^2 + \nu \left\| \nabla\mathbf{X}_i^{t+1} - \nabla\mathbf{X}_A - \mathbf{Z}_i^t \right\|_F^2 \right). \tag{27}$$

With Parseval's theorem, we take the fast Fourier transformation (FFT) of every variable and set its first-order derivative with respect to $\mathbf{X}_i^{t+1}$ to zero to solve the optimal intermediate latent. Let $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote FFT and inverse FFT, and $\nabla_x = [1, -1]^{\mathrm{T}}$ and $\nabla_y = [1, -1]$ denote gradient kernels along the x- and y-directions, respectively; then, the solution to (27) is shown as in (28), shown at the bottom of the next page.

Note that (28) is the exact closed-form solution to (27) if and only if $a = 1$. However, using a closed-form solution directly for each iteration is very likely to trap in a trivial solution with $\mathbf{X} = \mathbf{X}_A$ and $\mathbf{Z} = \mathbf{0}$, so we add a controlling parameter a to avoid this situation, and it can be empirically set to a small positive value, e.g., 10^{-2} in our experiments.

C. Blur Kernel Estimation

As is mentioned in Section II, estimating blur kernel using gradient rather than intensity of $\mathbf{X}$ and $\mathbf{Y}$ yields a more accurate result, so we adopt (20) as the objective of blur kernel estimation step. Let $\mathbf{X}_i = \mathbf{X}_i^T$ denote the iterative result after T step. Similarly, the closed-form solution can be obtained as in (29), shown at the bottom of the next page.

D. Nonblind Deconvolution

After the estimation of the blur kernel, we perform nonblind deconvolution on the blurred image difference and its auxiliary band using the estimated blur kernel. The nonblind deconvolution method proposed by Whyte et al. [18] is introduced in our experiments, which uses L_1-norm as the regularization. As is mentioned in Section III, we first compute the blurred image difference $\mathbf{Y}_D$ given the blurred image $\mathbf{Y}$, the estimated kernel $\hat{\mathbf{K}}$, and the auxiliary band image $\mathbf{X}_A$ via the following equation:

$$\mathbf{Y}_D = \mathbf{Y} - \hat{\mathbf{K}} \otimes \mathbf{X}_A. \tag{30}$$

Specifically, the deconvolution objective is written as in (31). This objective can be optimized by solving L_1-constrained problem, and the estimated difference $\hat{\mathbf{X}}_D$ is obtained

$$\hat{\mathbf{X}}_D = \arg\min_{\mathbf{X}_D} \left(\left\| \mathbf{Y}_D - \mathbf{X}_D \otimes \hat{\mathbf{K}} \right\|_F^2 + \|\nabla\mathbf{X}_D\|_1 \right). \tag{31}$$

Finally, according to Property 2, the estimation of latent image $\hat{\mathbf{X}}$ is as follows. We clip the output to make it range in $[0, 1]$

$$\hat{\mathbf{X}} = \min\left(\max\left(\hat{\mathbf{X}}_D + \mathbf{X}_A, \mathbf{0}\right), \mathbf{1}\right). \tag{32}$$

E. Overall Algorithm

The overall procedure to deblur RSIs using PMP and ABGI priors is shown in Algorithm 1. We use the hyperparameter b to increasingly adjust the penalty factor ν in the inner iterations of intermediate latent estimation. For each inner iteration, it is multiplied by b to make the penalty of imposing $\mathbf{Z}_i^t$ to be close to $\nabla\tilde{\mathbf{X}}_i^t$.

V. Experiments and Results

A. Evaluation Metrics

Evaluation of image deblurring involves full-reference image quality assessment (FR-IQA) and no-reference image quality assessment (NR-IQA). FR-IQA compares the restored test image to its corresponding undistorted image as a reference and computes a score as a quantitative metric reflecting the similarity of the image pair. FR-IQA is capable to reflect how close the test image is to its reference image, but full access to the original undistorted image is required, which is usually difficult to obtain for real-world image restoration. On the contrary, NR-IQA takes only the test image as input and computes the quality score by analyzing its statistical features. The following IQAs are used for evaluation in our paper, and we evaluate our results by MATLAB official implementations of mentioned IQAs.

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Algorithm 1 PMP + ABGI Deblurring

Input: Blurred image $\mathbf{Y}$, auxiliary image $\mathbf{X}_\mathrm{A}$.
Output: Estimated latent image $\hat{\mathbf{X}}$, estimated kernel $\hat{\mathbf{K}}$.

Initialize the kernel $\mathbf{K}_0$.
$\mathbf{X}_0 \leftarrow \mathbf{Y}$.
for $i = 0 \rightarrow I - 1$ **do**
  $\nu \leftarrow \nu_0$
  while $\nu \leq \nu_{\max}$ **do**
    $\mathbf{X}_i^0 \leftarrow \mathbf{X}_i$.
    for $t = 0 \rightarrow T - 1$ **do**
      Use (22) to update $\mathbf{M}_i^t$.
      Use (23) to update $\tilde{\mathbf{X}}_i^t$.
      Use (26) to update $\mathbf{Z}_i^t$.
      Use (28) to update $\mathbf{X}_i^{t+1}$.
    end for
    $\mathbf{X}_i \leftarrow \mathbf{X}_i^T$.
    $\nu \leftarrow b \cdot \nu$.
  end while
  Use (29) to update $\mathbf{K}_{i+1}$.
end for
$\hat{\mathbf{K}} \leftarrow \mathbf{K}_I$.
$\mathbf{Y}_\mathrm{D} \leftarrow \mathbf{Y} - \hat{\mathbf{K}} \otimes \mathbf{X}_\mathrm{A}$.
Solve (31) to obtain $\hat{\mathbf{X}}_\mathrm{D}$.
$\hat{\mathbf{X}} \leftarrow \min\big(\max\big(\hat{\mathbf{X}}_\mathrm{D} + \mathbf{X}_\mathrm{A}, \mathbf{0}\big), \mathbf{1}\big)$.

1) FR-IQA: FR-IQA includes peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) index [51]. Given a test image $\mathbf{X}_t$ and a reference $\mathbf{X}_r$ with the same size $m \times n$, the PSNR is computed as follows:

$$\mathrm{PSNR}(\mathbf{X}_t, \mathbf{X}_r) = 10 \log_{10} \frac{mn X_{\max}^2}{\sum_{(x,y)} (\mathbf{X}_t(x, y) - \mathbf{X}_r(x, y))^2} \tag{33}$$

where $X_{\max}$ denotes the possible maximum intensity of the image, e.g., $X_{\max}$ takes $2^B - 1$ for an image with B-bit depth and 1 for a floating point image ranged from 0 to 1.

PSNR is computed based on the mean square error of corresponding pixels, but it measures only numerical difference rather than the structural difference between test and reference images. To address this problem, the SSIM index is proposed. SSIM index [51] divides the two images into local patches and compares patch pairs in aspects of luminance, contrast, and structure, quantifies them to three factors, respectively, $l(x, y)$, $c(x, y)$, and $s(x, y)$ for patch pair x and y, and computes the local SSIM score by multiplication. The overall SSIM index is the mean SSIM index of local patches. Specifically, the local SSIM index is given as the following equation, where m_t and m_r denote the local mean value of test and reference, σ_t and σ_r denote the local standard deviation, σ_tr is the local cross covariance, and C_1 and C_2 are regularization constants:

$$\mathrm{SSIM}(x, y) = \frac{(2m_t m_r + C_1)(2\sigma_\mathrm{tr} + C_2)}{(m_t^2 + m_r^2 + C_1)(\sigma_t^2 + \sigma_r^2 + C_2)}. \tag{34}$$

2) NR-IQA: No-reference IQAs assess test images mainly by exploiting and quantifying statistical features of natural images and distortions. The blind/referenceless image spatial quality evaluator (BRISQUE) [52] utilizes the Gaussian-like empirical distribution of locally normalized luminance coefficients in the spatial domain of natural images and builds a natural scene statistic model for assessment. Unlike previous NR-IQAs, BRISQUE only quantifies the naturalness of an image as its score and does not reflect what category of distortion the image possesses.

Similarly, the natural image quality evaluator (NIQE) [53] is also based on natural scene statistics. NIQE requires neither human opinion scores nor prior knowledge of distortions and, therefore, does not need the involvement of subject opinion scores during model training.

The perception-based image quality evaluator (PIQE) [54] also extracts local features for quality quantification and gives a patch-level distortion classification as a basis for quality score assignment. Nevertheless, it focuses only on perceptual features in the spatial domain, so the assessment result of PIQE corresponds more to the human vision system.

For all three NR-IQAs, lower scores given by them indicate higher image quality and fewer distortions. BRISQUE and NIQE require training a model, while PIQE does not, and we apply the default BRISQUE/NIQE model provided by MATLAB's official implementation in our evaluation.

B. Data Preparation and Experiment Setup

1) Data Preparation: In our experiments, we evaluate our method on both open RSI datasets and GFDM (meaning high-resolution multimodal) RSIs. Basic information of the datasets is listed in Table I. The datasets are various in aspects, such as geographic resolutions, spectral types, sensor types, and carrier types.

We choose Pavia, PaviaU, airborne visible infrared imaging spectrometer (AVIRIS), and Sentinel-2 for open datasets. Pavia (available at https://www.ehu.eus/ccwintco/uploads/e/e3/Pavia.mat) and PaviaU (available at https://www.ehu.eus/ccwintco/uploads/e/ee/PaviaU.mat) datasets record scenes of the center and the University of Pavia, northern Italy. They have a geometric resolution of 1.3 m and contain 102 and 103 spectral bands, respectively. After discarding null samples from the original data, the Pavia dataset is 715 × 1096, and the size of PaviaU dataset is 340 × 610.

$$\mathbf{X}_i^{t+1} = \mathcal{F}^{-1}\left(\frac{\overline{\mathcal{F}(\mathbf{K}_i)} \odot \mathcal{F}(\mathbf{Y}) + \nu\left(\overline{\mathcal{F}(\nabla_x)} \odot \mathcal{F}(\nabla_x \otimes a\mathbf{X}_A + \mathbf{Z}_i^t) + \overline{\mathcal{F}(\nabla_y)} \odot \mathcal{F}(\nabla_y \otimes a\mathbf{X}_A + \mathbf{Z}_i^t)\right)}{\overline{\mathcal{F}(\mathbf{K}_i)} \odot \mathcal{F}(\mathbf{K}_i) + \nu\left(\overline{\mathcal{F}(\nabla_x)} \odot \mathcal{F}(\nabla_x) + \overline{\mathcal{F}(\nabla_y)} \odot \mathcal{F}(\nabla_y)\right)}\right) \tag{28}$$

$$\mathbf{K}_{i+1} = \mathcal{F}^{-1}\left(\frac{\overline{\mathcal{F}(\nabla_x \otimes \mathbf{X}_i)} \odot \mathcal{F}(\nabla_x \otimes \mathbf{Y}) + \overline{\mathcal{F}(\nabla_y \otimes \mathbf{X}_i)} \odot \mathcal{F}(\nabla_y \otimes \mathbf{Y})}{\overline{\mathcal{F}(\nabla_x \otimes \mathbf{X}_i)} \odot \mathcal{F}(\nabla_x \otimes \mathbf{X}_i) + \overline{\mathcal{F}(\nabla_y \otimes \mathbf{X}_i)} \odot \mathcal{F}(\nabla_y \otimes \mathbf{X}_i) + \lambda}\right) \tag{29}$$

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TABLE I

BASIC OVERVIEW OF THE RSI DATASETS, INCLUDING THE GEOGRAPHIC RESOLUTION, THE BAND NUMBER, THE SENSOR, AND CARRIER TYPE

Dataset Name	Geographic Resolution	Spectral Type	Total Band Number	Sensor	Carrier
Pavia/PaviaU	High (1.3m)	Hyperspectral	102/103	ROSIS	Aircraft
AVIRIS	Medium (15m)	Hyperspectral	224	AVIRIS	Aircraft
Sentinel-2	Medium (10m)	Multispectral	13	MSI	Sentinel-2A/2B Satellites
GFDM	High (2m)	Multispectral	8	TDICCD Image Sensors	GFDM Satellite

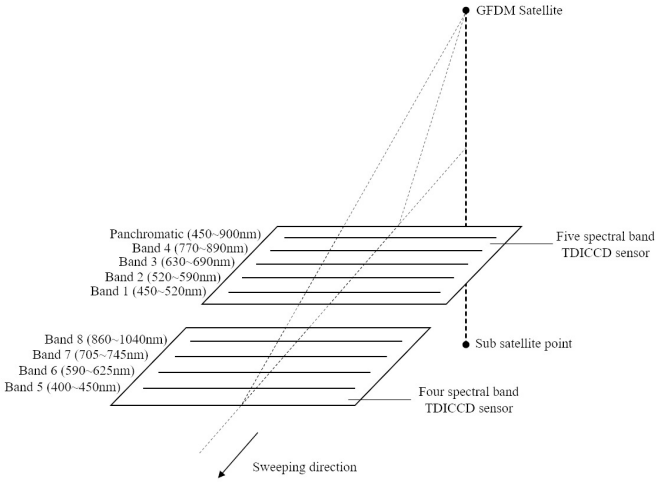


Fig. 2. Structure and spectral band distribution of TDICCD sensors carried by GFDM satellite.

We acquire AVIRIS scene named "f201015t01p00r15" from AVIRIS Data Portal (https://aviris.jpl.nasa.gov/dataportal/), which contains 224 spectral bands in total, and we crop a representative region, including farmlands, rivers, and urban areas with size 600 × 600. Sentinel-2 is an earth observation mission with a free and open data policy lead by the European Space Agency. The Sentinel-2 multispectral RSIs contain 13 bands in total, and the geographic resolution ranges from 10 to 60 m. In our experiments, we select four bands, including blue, green, red, and near infrared (NIR) at 10-m geographic resolution. The multispectral image data are available at Copernicus Open Access Hub (https://scihub.copernicus.eu/dhus/#/home), and four scenes are selected and cropped to 800 × 800 size.

GFDM RSIs are acquired by the GFDM satellite. It has one panchromatic and eight multispectral bands in total and has the highest geographic resolution at 0.5 m for the panchromatic band and 2 m for multispectral bands among all Chinese commercial satellites so far. The structure of the GFDM satellite and detailed band information is shown in Fig. 2. GFDM carries two time delay integration charge-coupled device (TDICCD) image sensors and acquires ground electromagnetic reflection by sweeping along its flying direction. The first sensor collects five spectral bands, including the panchromatic band and bands 1–4, and the second collects bands 5–8. During the collection, the first TDICCD sensor is not affected and receives clear images, while the second TDICCD sensor is not stably set up and collects blurred images of bands 5–7. In this situation, we may utilize clear images from the first sensor to recover blurred images from the second sensor, which coincides with the situation of ABGI prior. In our experiments, we select eight scenes from GFDM RSIs and crop them to size 800 × 800. The detailed scene description is shown in Table II and Fig. 3.

TABLE II

SCENE DESCRIPTION OF GFDM RSIS

Index	Date(yyyy/mm/dd)	Description
GFDM01	2021/09/01	Deserts, lakes
GFDM02	2021/10/17	Farmlands, canals, towns, boats
GFDM03	2022/01/08	Mountains, iced rivers, snowfields
GFDM04	2022/01/16	Glaciers, wasteland
GFDM05	2022/02/14	Farmlands, rivers, road bridges, flyovers
GFDM06	2022/03/01	Hills, farmlands, roads, villages
GFDM07	2022/04/02	Mountains, towns, forests
GFDM08	2022/04/26	Urban buildings and roads

2) Experiment Setup: We compare our method with several competitive blind image deblurring methods, including DCP [27], ECP [28], and PMP [22]. We evaluate both simulated and real-world blur. According to [55], we test three different blur types in the simulated experiments: Gaussian blur to simulate atmospheric disturbance, motion blur to simulate platform instability, and disk blur to simulate camera defocus. For Pavia, PaviaU, and AVIRIS datasets, we select band 20 to simulate blur and use band 50 as the auxiliary band; for Sentinel-2 and GFDM images, we select band 2 (green, spectral band ranged at 520–590 nm) to simulate blur and use band 3 (red, spectral band ranged at 630–690 nm) as the auxiliary band. Five image quality metrics are used to assess the deblurring quality of methods, including two full-reference metrics (PSNR and SSIM) and three no-reference metrics (BRISQUE, NIQE, and PIQE). In the real-world experiments, we evaluate on real blur of GFDM band 7 (spectral band ranged at 705–745 nm) and use band 4 (spectral band ranged at 770–890 nm) as the auxiliary band. Since the ground truth of real-world blur is not available, only no-reference metrics are introduced for quality assessment.

Our experiments are based on the MATLAB implementation of [22] and run on a personal laptop with i7-7700HQ and 16-GB RAM. For parameter setting, we set our parameters of kernel estimation and deconvolution as follows: $\mu = 4 \times 10^{-3}$, $\nu_0 = 8 \times 10^{-3}$, $\nu_{\max} = 10^5$, $a = 10^{-2}$, $b = 2$, $\kappa = 0.1$, $I = 5$, and $T = 3$. The source codes of the proposed method are provided at https://github.com/abgi2023/abgi.

C. Experimental Results

In our experiments, we compare the quantitative performance of our deblurring method with some competitive blind deblurring methods, including DCP [27], ECP [28], and PMP [22]. Two full-reference metrics, namely, PSNR and SSIM, and three no-reference metrics, namely, BRISQUE, NIQE, and PIQE, are involved. No-reference metrics are introduced for being good references to reflect clearness or naturalness for real-world deblurring evaluation. Note that higher deblurring quality gives higher PSNR and SSIM scores and lower

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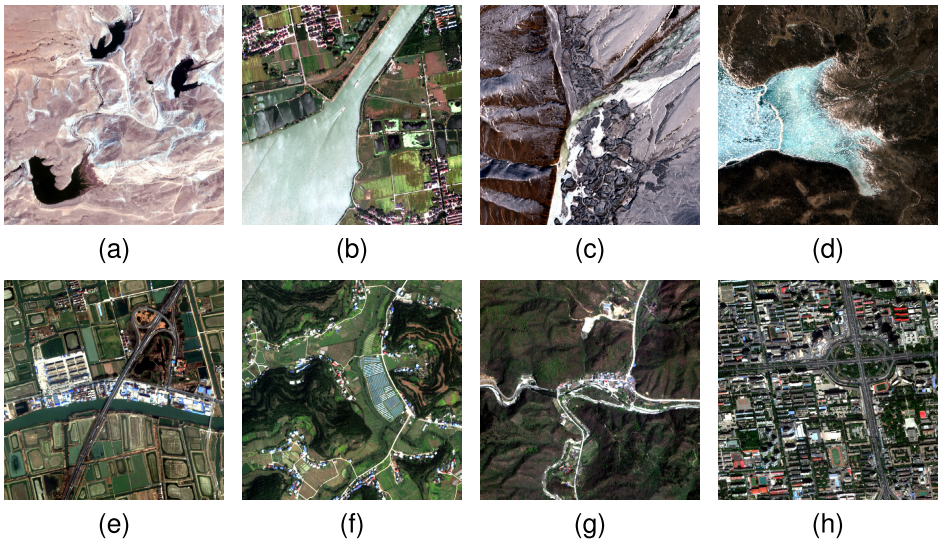


Fig. 3. Overview of eight GFDM scenes in RGB. Bands 1–3 of GFDM images correspond to blue, green, and red channels, respectively. (a) GFDM01. (b) GFDM02. (c) GFDM03. (d) GFDM04. (e) GFDM05. (f) GFDM06. (g) GFDM07. (h) GFDM08. Zoom in to view details.

BRISQUE, NIQE, and PIQE scores. For the simulated blur, we test Gaussian blur with $\sigma = 3$, motion blur with direction at 45°, and line length at 20, and use a disk with radius $r = 7$ to simulate defocus blur; thus, the valid size of all kernels is approximately 15, and kernels are normalized.

According to the spectral type, Pavia, PaviaU, and AVIRIS will be categorized to hyperspectral datasets, and Sentinel-2 and GFDM to multispectral datasets. Similarly, Pavia, PaviaU, and GFDM are categorized to high resolution, and AVIRIS and Sentinel-2 are categorized to medium resolution according to the geographic resolution. The following experiments exhibit superior performance of the proposed method despite the spectral type or the geographic resolution of the RSIs.

1) Results on Pavia and PaviaU Datasets: The Pavia and PaviaU datasets are high resolution and hyperspectral. The quantitative deblurring results for Pavia and PaviaU datasets are shown in Table III. It is shown that our method has superior performance on three simulated blur types than all other comparative methods. Fig. 4 shows deblurring results in the simulated Gaussian blur case. None of the comparative methods but ours can recover clear edges of the building from heavy Gaussian blur. Detailed results in the motion blur case are also shown in Fig. 5. We select a building with a clear sign for comparison, and it is shown that our method recovers a clear sign with the help of the auxiliary band.

2) Results on AVIRIS Dataset: The AVIRIS dataset is medium resolution and hyperspectral. The quantitative results for AVIRIS dataset are shown in Table IV. Evaluation on AVIRIS is a little challenging, since its geographic resolution at around 15 m is far behind those of Pavia, PaviaU, and GFDM images. Our method shows slightly better performance than other methods. Detailed deblurring results on defocus blur are also shown in Fig. 6. Defocus blur can cause great damage to texture information; however, with the introduction of auxiliary band, it is possible to recover clear texture.

3) Results on Multispectral Datasets: The Sentinel-2 dataset is medium resolution and multispectral. The quantitative average results of Sentinel-2 multispectral images are shown in Table V. Although the geographic resolution of 10 m is far behind those of Pavia, PaviaU, and GFDM images, our method still shows superior performance

TABLE III

AVERAGE QUANTITATIVE DEBLURRING PERFORMANCE ON PAVIA AND PAVIAU DATASETS. NOTE THAT HIGHER PSNR/SSIM AND LOWER BRISQUE/NIQE/PIQE INDICATE BETTER PERFORMANCE

Blur type	Metric	DCP [27]	ECP [28]	PMP [22]	Ours
Gaussian ($\sigma = 3$)	PSNR	19.21	19.20	19.19	**23.64**
	SSIM	0.4879	0.4822	0.4645	**0.8364**
	BRISQUE	43.50	43.49	43.54	**33.47**
	NIQE	7.72	8.02	8.05	**3.71**
	PIQE	75.09	72.81	71.86	**20.14**
Motion (45°, length=20)	PSNR	21.65	21.68	22.54	**25.08**
	SSIM	0.6717	0.6754	0.6896	**0.8545**
	BRISQUE	40.62	40.94	40.16	**30.49**
	NIQE	7.72	7.55	7.12	**4.14**
	PIQE	26.40	26.88	28.65	**18.95**
Defocus (r = 7)	PSNR	20.61	20.63	20.64	**24.15**
	SSIM	0.5738	0.5750	0.5768	**0.8482**
	BRISQUE	45.76	45.42	46.25	**30.33**
	NIQE	6.75	6.89	6.94	**3.70**
	PIQE	70.78	69.47	68.53	**19.77**

TABLE IV

QUANTITATIVE DEBLURRING PERFORMANCE ON AVIRIS DATASET. NOTE THAT HIGHER PSNR/SSIM AND LOWER BRISQUE/NIQE/PIQE INDICATE BETTER PERFORMANCE

Blur type	Metric	DCP [27]	ECP [28]	PMP [22]	Ours
Gaussian ($\sigma = 3$)	PSNR	19.18	19.17	19.03	**19.94**
	SSIM	0.4089	0.4073	0.3995	**0.6073**
	BRISQUE	50.82	50.48	50.13	**9.32**
	NIQE	9.00	9.95	10.65	**3.58**
	PIQE	48.58	45.12	40.69	**33.69**
Motion (45°, length=20)	PSNR	20.35	20.40	20.45	**20.87**
	SSIM	0.6436	0.6547	0.6467	**0.7123**
	BRISQUE	43.43	43.42	43.41	**15.47**
	NIQE	7.85	7.39	7.45	**4.10**
	PIQE	43.70	45.41	45.32	**32.44**
Defocus ($r = 7$)	PSNR	19.73	19.73	19.71	**20.26**
	SSIM	0.4733	0.4733	0.4726	**0.6287**
	BRISQUE	54.24	53.79	53.62	**10.23**
	NIQE	6.82	7.04	7.38	**3.71**
	PIQE	53.53	54.95	50.71	**33.48**

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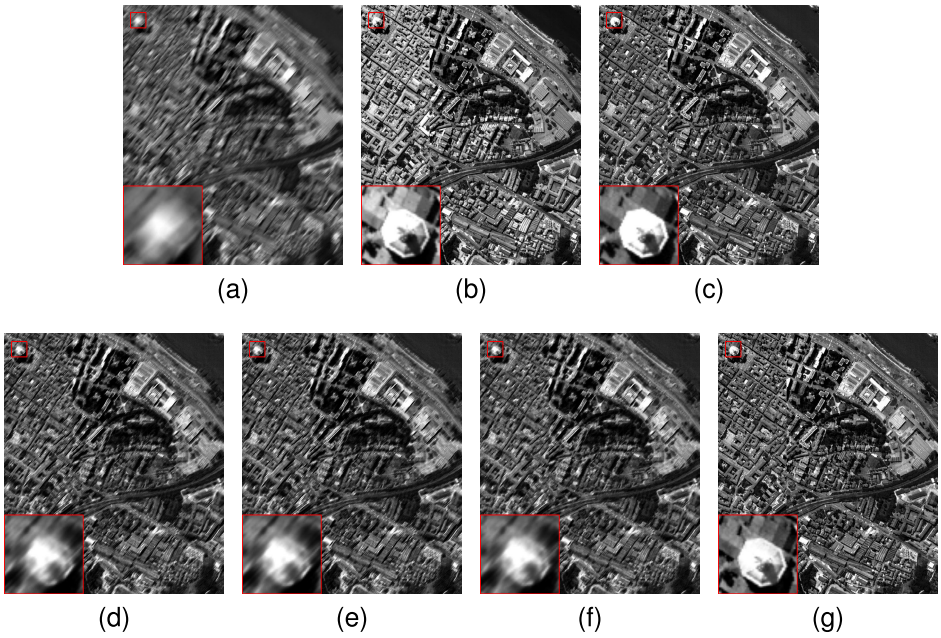


Fig. 4. Deblurring result comparison on Pavia dataset. (a) Blurred ground truth. (b) Auxiliary band. (c) Ground truth. (d) Result of DCP. (e) Result of ECP. (f) Result of PMP. (g) Our result. Zoom in to view details.

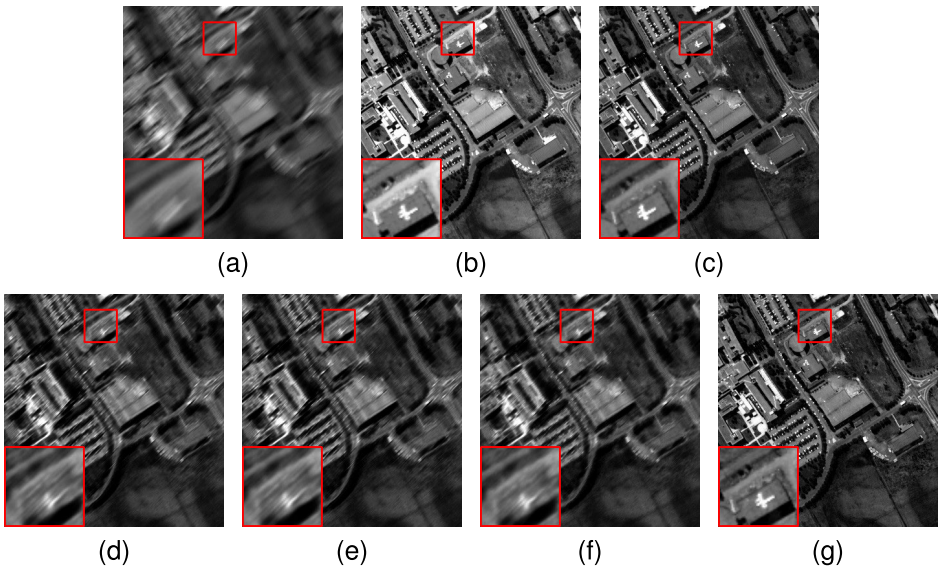


Fig. 5. Deblurring result comparison on PaviaU dataset. (a) Blurred ground truth. (b) Auxiliary band. (c) Ground truth. (d) Result of DCP. (e) Result of ECP. (f) Result of PMP. (g) Our result. Zoom in to view details.

than other methods. For simulated motion blur, our method performs much better than others except the PIQE metric. The sensitivity to slight ringing artifacts results in inferior PIQE scores. Detailed deblurring results on defocus blur are also shown in Fig. 7. Defocus blur can cause great damage to texture information, but it is possible to recover clear texture with the introduction of the auxiliary band.

4) Results on GFDM Images: The GFDM dataset is high resolution and multispectral. The average quantitative results for eight GFDM images are shown in Table VI. Our method outperforms others except the PIQE metric in motion blur scenes for the same reason mentioned in PaviaU dataset. Besides simulated blur, real-world blur on GFDM images is also evaluated. Our method outperforms both simulated blur and real-world blur. Detailed deblurring results for simulated blur are also shown. In GFDM02 (see Fig. 8), the road lying on the farmland is recovered clearly, while others failed; in GFDM05 (see Fig. 9), the shadows and edges of buildings are recovered clearly; in GFDM06 (see Fig. 10), textures of farmland are completely recovered despite of serious defocus blur. For real-world results on GFDM images, detailed results of PMP and our method are shown in Figs. 11 and 12. For most of the listed scenes, deblurring results of PMP still remain slight blur or suffer texture deformation, while ABGI-based method yields clear outputs.

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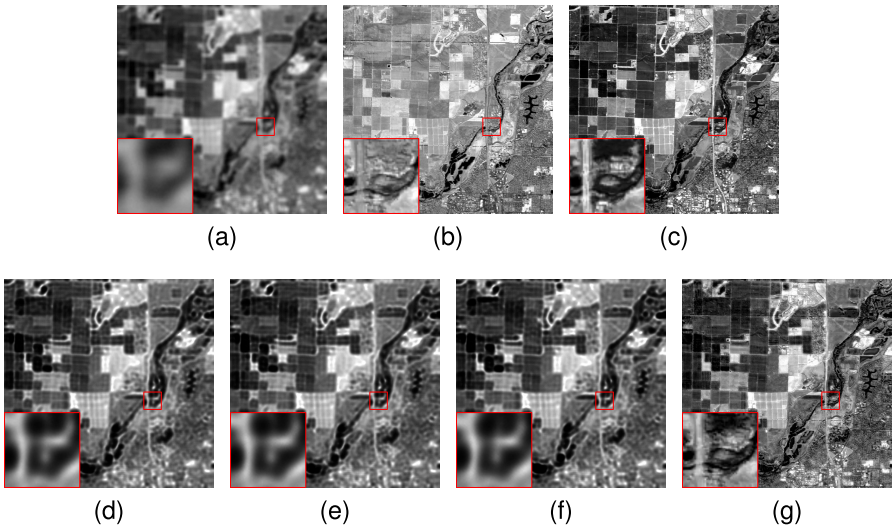


Fig. 6. Deblurring result comparison on AVIRIS dataset. (a) Blurred ground truth. (b) Auxiliary band. (c) Ground truth. (d) Result of DCP. (e) Result of ECP. (f) Result of PMP. (g) Our result. Zoom in to view details.

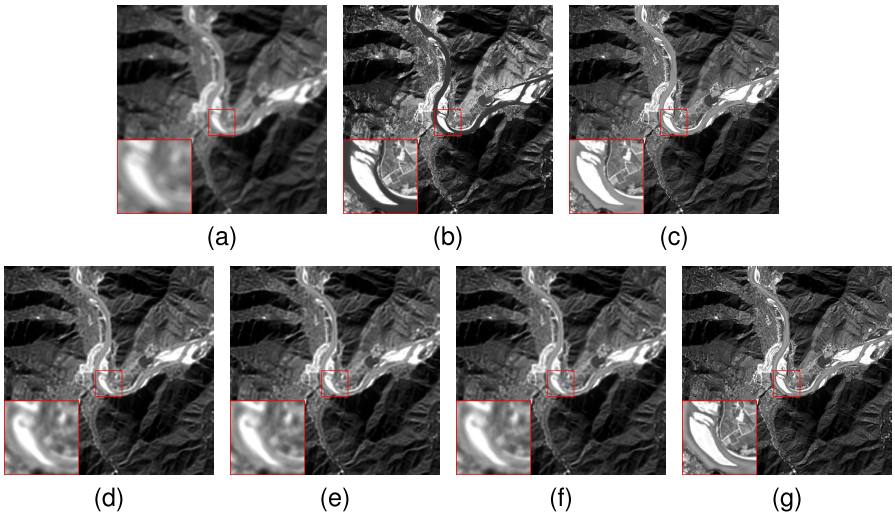


Fig. 7. Deblurring result comparison on Sentinel-2 multispectral images. (a) Blurred ground truth. (b) Auxiliary band. (c) Ground truth. (d) Result of DCP. (e) Result of ECP. (f) Result of PMP. (g) Our result. Zoom in to view details.

TABLE V

QUANTITATIVE DEBLURRING PERFORMANCE ON SENTINEL-2 MULTISPECTRAL IMAGES. NOTE THAT HIGHER PSNR/SSIM AND LOWER BRISQUE/NIQE/PIQE INDICATE BETTER PERFORMANCE

Blur type	Metric	DCP [27]	ECP [28]	PMP [22]	Ours
Gaussian ($\sigma = 3$)	PSNR	22.61	22.60	22.27	**30.50**
	SSIM	0.6370	0.6370	0.6283	**0.9085**
	BRISQUE	50.04	49.66	49.57	**19.64**
	NIQE	7.24	7.73	8.05	**3.03**
	PIQE	44.33	44.02	38.91	**14.20**
Motion (45°, length=20)	PSNR	22.08	22.07	22.70	**30.81**
	SSIM	0.6485	0.6498	0.6832	**0.8942**
	BRISQUE	38.09	37.93	34.64	**22.16**
	NIQE	8.57	8.66	8.44	**4.09**
	PIQE	9.77	9.76	**9.47**	15.54
Defocus (r = 7)	PSNR	21.78	21.69	21.71	**29.81**
	SSIM	0.5936	0.5908	0.5921	**0.9031**
	BRISQUE	50.78	50.81	49.02	**26.09**
	NIQE	6.53	7.01	7.24	**3.03**
	PIQE	38.73	37.97	34.27	**15.10**

VI. ANALYSIS AND DISCUSSION

A. Effectiveness of PMP and ABGI

To verify the effectiveness of PMP prior, we calculate the histogram of clear and blurred image PMP, respectively. The real-world blur kernel estimated from bands 6 and 7 is used to simulate blurring, and all clear bands (i.e., bands 1–4 and 8) of five scenes are taken into account. We first count the PMP of each clear band and then convolve it with the kernel to obtain the blurred band and count its PMP. The histogram is shown in Fig. 13. It is obvious that smaller PMP values tend to appear on clear images, and greater PMP values tend to appear on blurred images. In clear images, PMP values concentrate at smaller values near zero and have lower variance than those in blurred images. Based on this, the sparsity of PMP is shown in clear images, and the effectiveness of PMP as priors is guaranteed.

We compare ABGI prior with the frequently used intensity gradient prior. For intensity gradient prior, we calculate the absolute value of gradient at every pixel; for ABGI prior, we select two clear bands, obtain the gradient, respectively, and calculate the difference of band gradients. The histogram of absolute value is shown in Fig. 14. The values of ABGI are more likely to be smaller than those of the intensity gradient prior, indicating that the ABGI prior is more effective than the intensity gradient prior.

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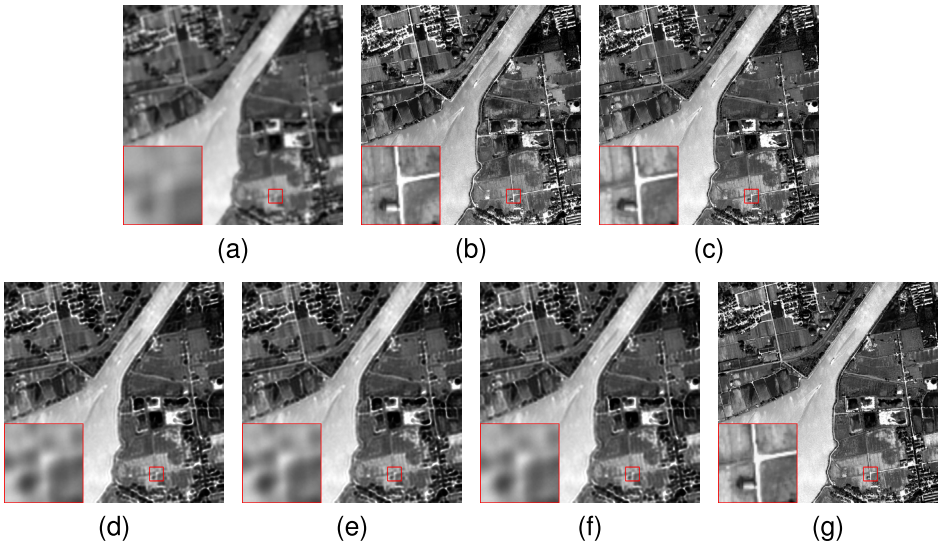


Fig. 8. Simulated deblurring result comparison on GFDM02. (a) Blurred ground truth. (b) Auxiliary band. (c) Ground truth. (d) Result of DCP. (e) Result of ECP. (f) Result of PMP. (g) Our result. Zoom in to view details.

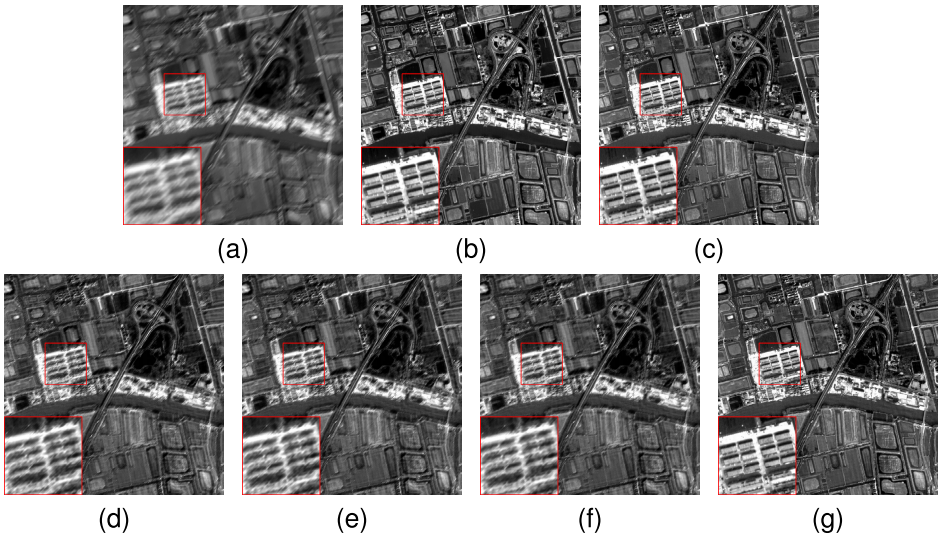


Fig. 9. Simulated deblurring result comparison on GFDM05. (a) Blurred ground truth. (b) Auxiliary band. (c) Ground truth. (d) Result of DCP. (e) Result of ECP. (f) Result of PMP. (g) Our result. Zoom in to view details.

B. Selection of Auxiliary Band

Since the ABGI prior substantially improves deblurring performance, choosing the appropriate auxiliary band is significant. Intuitively, the intensity at some spectral bands is close to that of its neighbor spectral band for many cases, so we may assume that closer spectral band yields better performance as an auxiliary band.

We verify this assumption on both simulated and real-world blur. For the simulated blur, we select band 21 of PaviaU and select some of its neighbor spectral bands (including itself) as auxiliary bands. Quantitative performances are shown in Fig. 15, and the wavelength increases uniformly, as the band number grows. For the real-world blur, we select bands 6 and 7 of GFDM images as real blur scenes and bands 1–4 and 8 as auxiliary bands. The wavelength range increases from bands 1–4 to 8. Band 6 lies between bands 2 and 3, and band 7 lies between bands 3 and 4. No-reference metrics of the real-world blur are shown in Fig. 16. It is empirically shown that closer auxiliary bands are more likely to have superior performance. Note that the green lines denote the comparative result without ABGI prior, and all results with ABGI prior have superior quantitative performance.

C. Computational Complexity and Running Time

The computational complexity of the proposed algorithm is as follows. The algorithm consists of kernel estimation and deconvolution. For the kernel estimation, three layers of loops are contained and the update formulas inside cost $\mathcal{O}(mn\log(mn))$ steps because of FFT operations, where $m \times n$ is the size of the image, so the total computational complexity is $\mathcal{O}(I \cdot \log(\nu_{\max}/\nu_0) \cdot T \cdot mn\log(mn))$. For deconvolution, the computational complexity is around $\mathcal{O}(mn)$, which is much smaller than that of kernel estimation. Therefore, the

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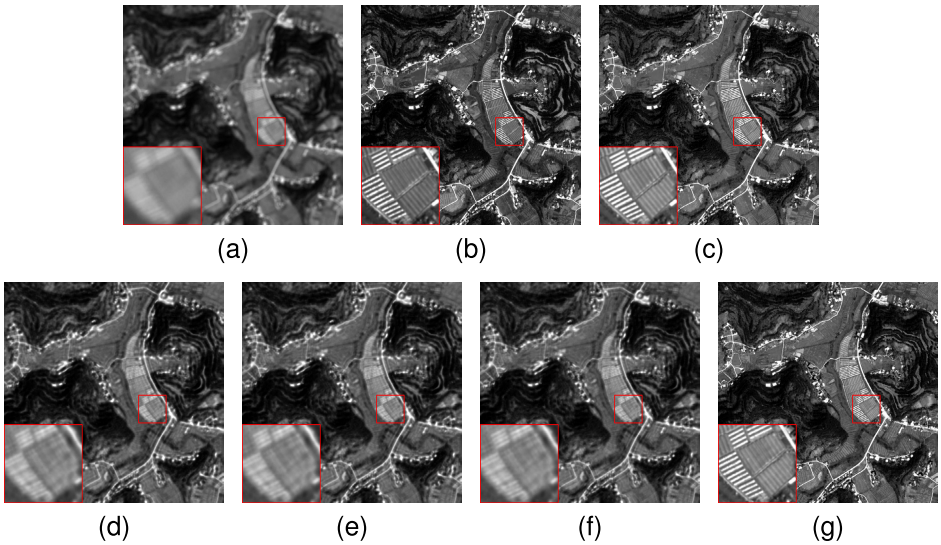


Fig. 10. Simulated deblurring result comparison on GFDM06. (a) Blurred ground truth. (b) Auxiliary band. (c) Ground truth. (d) Result of DCP. (e) Result of ECP. (f) Result of PMP. (g) Our result. Zoom in to view details.

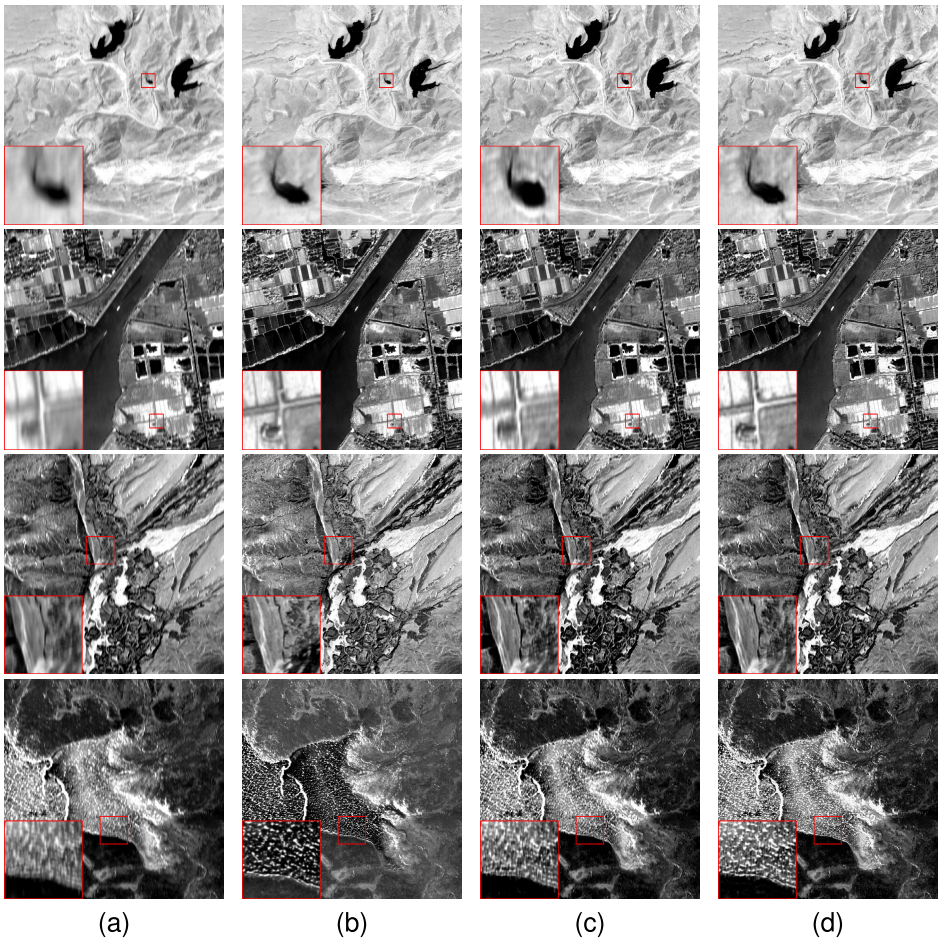


Fig. 11. Real-world deblurring result comparison on GFDM images. The four rows refer to GFDM01, GFDM02, GFDM03, and GFDM04, respectively. (a) Blurred band. (b) Auxiliary band. (c) Result of PMP. (d) Our result. Zoom in to view details.

overall computational complexity is $\mathcal{O}(I \cdot \log(v_{\max}/v_0) \cdot T \cdot mn \log(mn))$.

We compare the running time of all methods on three different sizes, namely, 400 × 400, 600 × 600, and 800 × 800. We discover that finding minimal pixels becomes a program hotspot (i.e., code segments that cost most of time and resources), so we substitute this part with a C language implementation supported by MATLAB MEX platform to boost the process. Detailed running time comparison is shown in Table VII. Note that for PMP, we list both pure MATLAB

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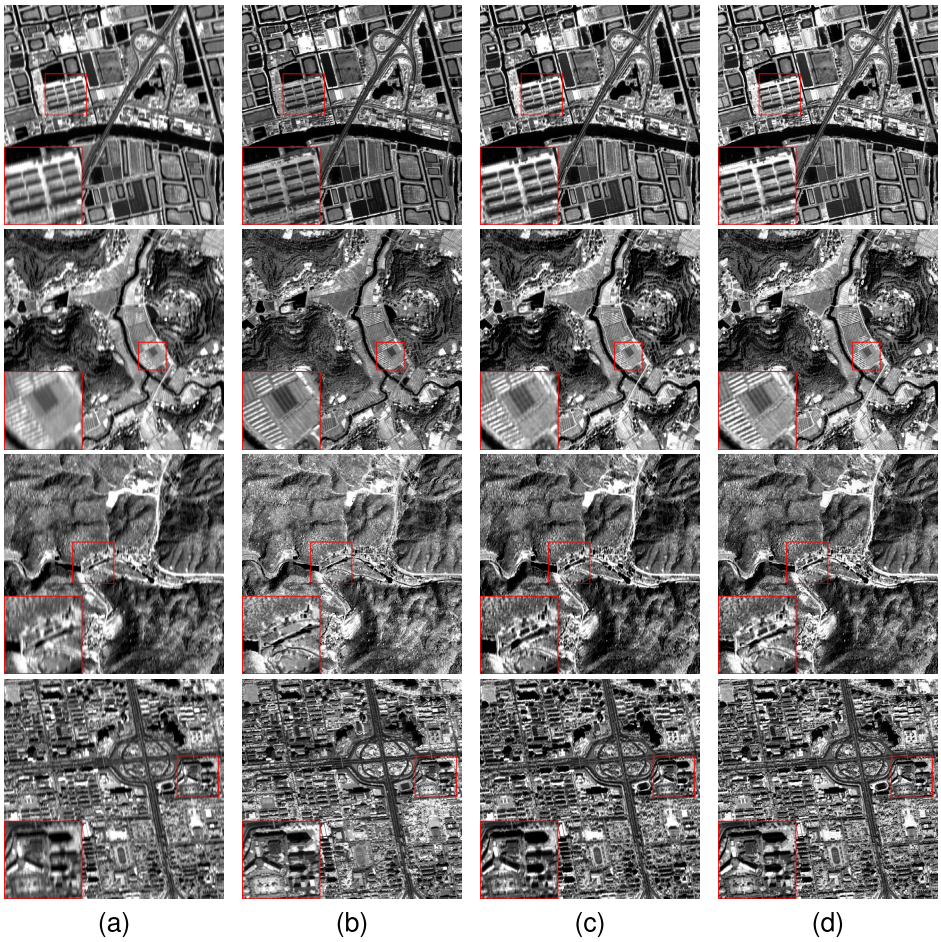


Fig. 12. Real-world deblurring result comparison on GFDM images. The four rows refer to GFDM05, GFDM06, GFDM07, and GFDM08, respectively. (a) Blurred band. (b) Auxiliary band. (c) Result of PMP. (d) Our result. Zoom in to view details.

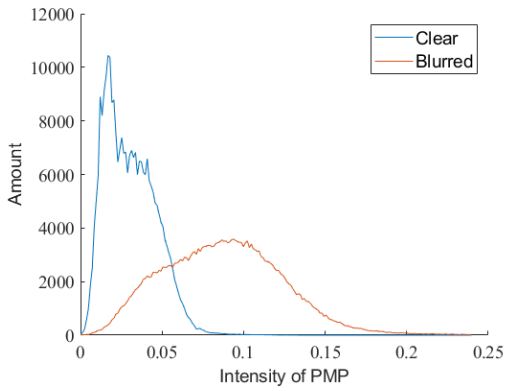


Fig. 13. Histogram of clear and blurred image PMP. The x-axis represents the value PMP takes, and the y-axis represents the frequency.

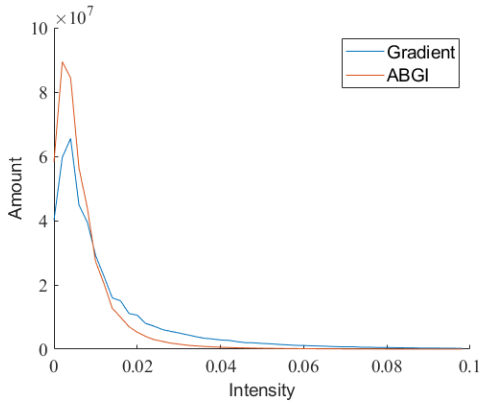


Fig. 14. Absolute value histogram of ABGI and intensity gradient prior. The x-axis represents the value, and the y-axis represents the frequency.

and MATLAB with C version for fairness. The involvement of ABGI prior increases only a few seconds, and it dramatically increases the deblurring performance.

D. Large-Scale Application

Deblurring applications on large-scale RSIs is an important yet hardly discussed issue in previous works. Large-scale RSIs have a much greater size than dataset images. The typical height/width of a large-scale RSI scene is often around 8000–10 000 pixels, and the quadratic increasing speed of processing time cost is unacceptable. For our method, we adopt a time-saving strategy to dramatically cut down the computational cost yet maintain the processing quality. Under the assumption that the blur effects are uniform and spatially invariant, a relatively small region is enough for kernel estimation.

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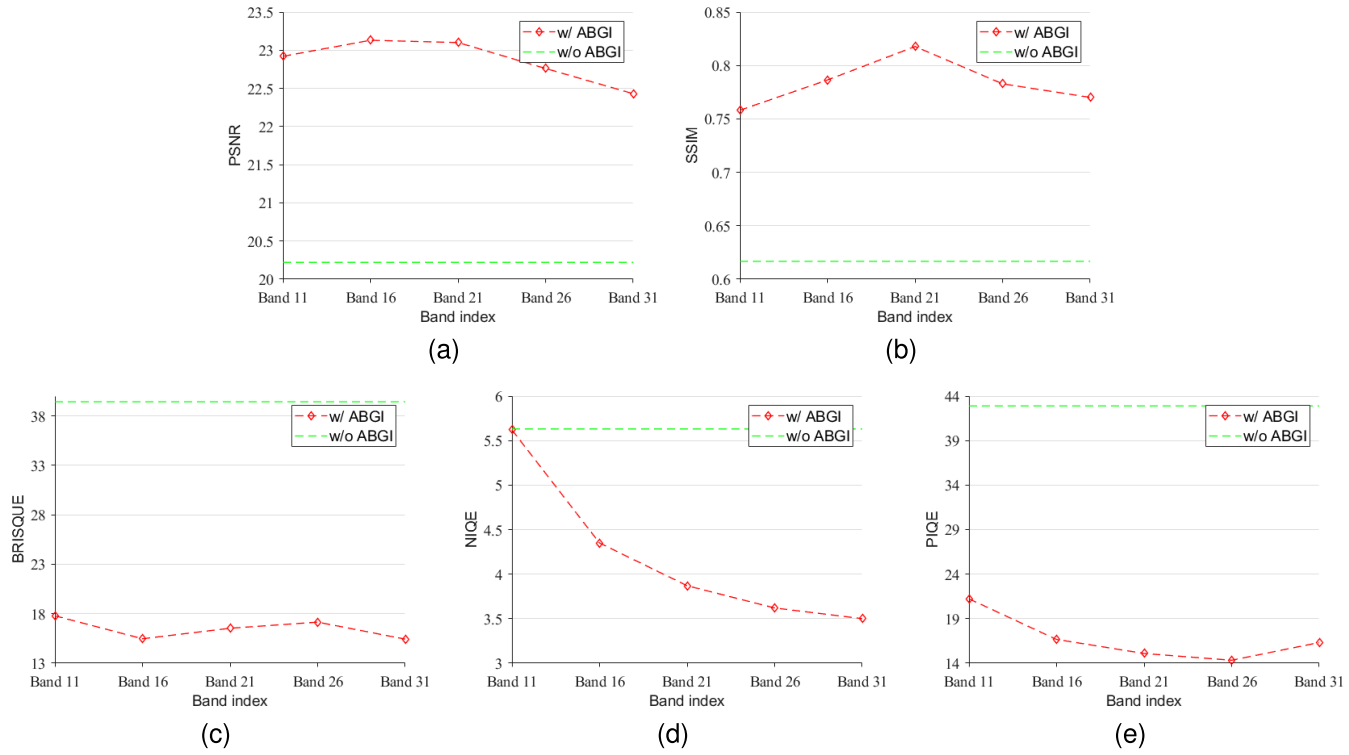


Fig. 15. Quantitative performance between different auxiliary bands on PaviaU dataset. Band 21 is selected to simulate the blur. (a) PSNR. (b) SSIM. (c) BRISQUE. (d) NIQE. (e) PIQE.

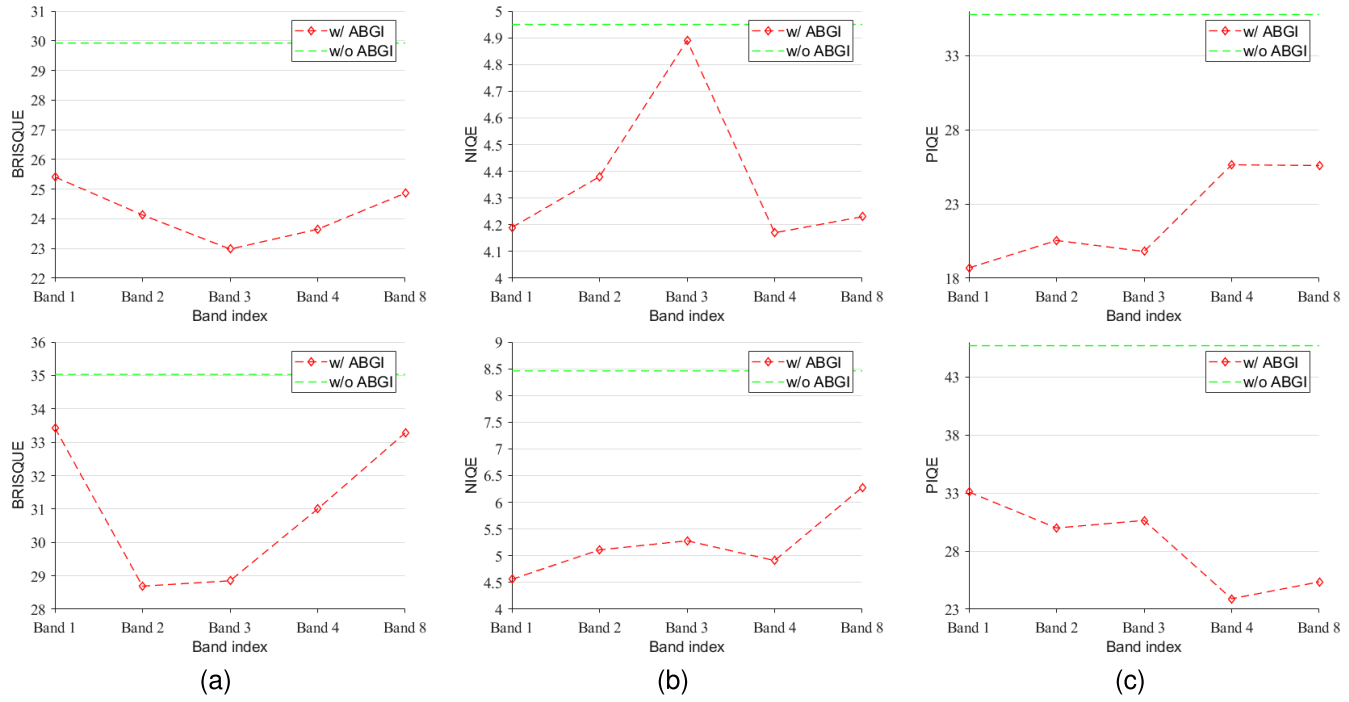


Fig. 16. Quantitative performance between different auxiliary bands on GFDM images. Bands 6 and 7 are real blurred. The first row relates to band 6, and the second row relates to band 7. (a) BRISQUE. (b) NIQE. (c) PIQE.

To be specific, we first divide the image into tiles, select a center tile to estimate the kernel instead of the whole image, and perform the deconvolution tilewise. Let M and N denote the size of the large-scale image and m and n denote the size of the center tile. Let $M = k_1 m$ and $N = k_2 n$, where k_1 and k_2 are ratios of long and short sides. From Section VI-C, we assume the number of basic operations on kernel estimation is $\alpha_1 mn \log(mn)$ and $\alpha_2 mn$ on deconvolution, where α_1 and α_2 are positive constant terms. If we use our method directly on the large-size image, the total amounts

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TABLE VI

AVERAGE QUANTITATIVE DEBLURRING PERFORMANCE ON GFDM IMAGES. NOTE THAT HIGHER PSNR/SSIM AND LOWER BRISQUE/NIQE/PIQE INDICATE BETTER PERFORMANCE

Blur type	Metric	DCP [27]	ECP [28]	PMP [22]	Ours
Gaussian ($\sigma = 3$)	PSNR	21.24	21.18	20.84	**29.34**
	SSIM	0.5278	0.5251	0.5062	**0.8939**
	BRISQUE	49.39	48.99	48.65	**24.24**
	NIQE	8.74	9.27	9.71	**3.30**
	PIQE	39.69	38.98	35.15	**20.84**
Motion (45°, length=20)	PSNR	22.40	22.48	22.79	**30.65**
	SSIM	0.6598	0.6657	0.6843	**0.9064**
	BRISQUE	40.35	40.18	37.78	**27.09**
	NIQE	8.08	8.06	7.54	**3.79**
	PIQE	**15.64**	16.17	17.27	17.13
Defocus ($r = 7$)	PSNR	21.91	21.77	21.96	**29.99**
	SSIM	0.5870	0.5819	0.5897	**0.9040**
	BRISQUE	52.80	52.53	51.60	**25.65**
	NIQE	7.05	7.59	7.66	**3.22**
	PIQE	44.61	43.90	40.19	**21.05**
Real world	BRISQUE	41.23	39.69	35.98	**34.35**
	NIQE	7.65	7.35	6.18	**4.40**
	PIQE	60.60	57.48	41.79	**23.58**

TABLE VII

RUNNING TIME COMPARISON IN SECONDS

Size	DCP	ECP	PMP	PMP (w/ optim.)	ABGI (w/ optim.)
400×400	363.4	373.3	20.50	13.54	14.33
600×600	793.7	828.0	41.54	33.16	34.80
800×800	1423	1339	71.59	60.88	62.26

of basic operations are relatively $\alpha_1 k_1 k_2\ mn \log(k_1 k_2\ mn)$ and $\alpha_2 k_1 k_2\ mn$. If we adopt the strategy, the amounts are, respectively, $\alpha_1 mn \log(mn)$ and $\alpha_2 mn \cdot (MN/mn) = \alpha_2 k_1 k_2\ mn$. The amounts of the deconvolution step operations are the same, but the kernel estimation step operations with the strategy are much fewer than those without the strategy. Adopting the strategy saves around $\alpha_1\ mn[(k_1 k_2 - 1)\log(mn) + k_1 k_2 \log(k_1 k_2)]$ basic operations and achieves $k_1 k_2[1 + \log(k_1 k_2)/\log(mn)]$ times speedup on the kernel estimation.

We also validate the processing quality of the full-size GFDM RSIs with the strategy. The eight GFDM images in the experiment section are cropped from their large-scale scenes. The maximum size of these images is 9754×9080, and the minimum size is 9458×9083. In practice, the average processing time with the strategy is around 1200 s, which is much less than the theoretical speed without the strategy (more than 9000 s). The quantitative results are listed in Table VIII, and we can see the strategy maintains almost the same processing quality compared with the results of the experiment section.

E. Limits

Our image deblurring method shows good performance. However, there are still some limits found in our method. First, the kernel estimation is not robust enough to various noises. For example, additive Gaussian noise with higher variance added to blurred images may cause unnatural kernel estimation. Band 5 of GFDM images is degraded simultaneously by blur, Gaussian, and strip noises, which severely interfere with blur kernel estimation. To solve this, joint approaches combining both denoising and deblurring are proposed by some former work. How to denoise and deblur simultaneously, effectively, and robustly can be considered in future works. Another limit is that imposing L_0-norm constraint on PMP is too radical for some scenes. Former work, including [22], uses L_0-norm to regularize PMP, because the zero elements of natural clear image PMP are usually dominant. However, RSIs are less likely to possess as many zero pixels as natural clear images. Using L_1-norm to regularize PMP is more appropriate, but L_1-regularized PMP is difficult to be imposed efficiently. Explicitly optimizing the L_1-regularized PMP term requires extra FFT and IFFT calculation and increases computational costs. Balancing both of them is worth discussing and solving in the future.

TABLE VIII

PROCESSING QUALITY COMPARISON OF THE CROPPED AND THE FULL-SIZE IMAGE. THE FULL-SIZE IMAGE IS PROCESSED USING THE TIME-SAVING STRATEGY. THE STRATEGY MAINTAINS ALMOST THE SAME PROCESSING QUALITY

Scene	Cropped Size (800×800)			Full Size		
	BRISQUE	NIQE	PIQE	BRISQUE	NIQE	PIQE
GFDM01	31.91	4.19	16.15	29.30	3.96	19.93
GFDM02	32.99	4.53	20.54	39.99	2.72	21.56
GFDM03	25.43	4.07	20.10	29.38	3.14	16.55
GFDM04	37.53	4.65	22.08	10.43	3.52	14.85
GFDM05	40.30	3.84	29.43	41.97	3.86	29.81
GFDM06	24.70	4.01	24.25	22.66	4.38	23.26
GFDM07	40.67	4.77	26.62	40.18	5.10	23.33
GFDM08	41.26	5.11	29.45	25.10	3.87	24.24

VII. CONCLUSION

We propose a novel prior for RSI deblurring called ABGI, which exploits gradient information similarity between different spectral bands. We show that by a simple subtract-then-add step, it is easy to utilize the ABGI prior to all gradient sparsity regularizations and, thus, to deblur multispectral or hyperspectral images. An improved overall procedure utilizing both PMP and ABGI priors is proposed. In our experiments, we use RSI datasets of different spectral types and geographic resolutions to estimate our method on both simulated and real-world blur scenes. We use multiple image quality metrics to quantify the deblurring performances, and our methods show superior performance than other state-of-the-art methods. The effectiveness of the proposed method, the selection of the auxiliary band, the large-scale application, the running time, and the limits are also discussed. In the future studies, we will focus more on improving robustness of RSI deblurring methods against various noises, such as non-Gaussian noises, strip noises, and so on. Effective and large-scale applications of joint approaches combining multiple restoring tasks are worth focusing on. How to select proper regularizations and how to impose the regularizations efficiently should be considered as well in our future research.

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